

MATH0110 Analysis for Joint Honours

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0110
<i>Level:</i>	4 (UG)
<i>Normal student group(s):</i>	UG: Y1 Mathematics programmes
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	1
<i>Assessment:</i>	70% examination, 20% coursework, 10% departmental test(s). The coursework is divided into 15% analysis coursework and 5% Proofs and Foundations coursework. The test is a midsessional exam in January. In order to pass the module you must have at least 40% for both the examination mark and the final weighted mark.
<i>Normal Pre-requisites:</i>	A* in A-level Mathematics and Further Mathematics
<i>Lecturer:</i>	Dr B Navarro Lamedá

Course Description and Objectives

This module is an introduction to mathematical analysis, one of the most important and well developed strands of pure mathematics with many elegant and beautiful theorems. The aim is to introduce students to the ideas of formal definitions and rigorous proofs (one of the fundamental features of modern mathematics, and something that is not familiar from A-level), and to develop their powers of logical thinking. This module leads on to second and third year modules in mathematical analysis and related areas. It is intended for first year students on the Mathematics and Statistical Science degree.

Recommended Texts

- Haggarty, *Fundamentals of Mathematical Analysis* 2nd ed (Pearson)
- Binmore, *Introduction to Mathematical Analysis* (CUP)
- M Spivak, *Calculus* (Publish or Perish)
- R Bartle and D Sherbert, *Introduction to Real Analysis* (Wiley)
- MH Protter and CB Morrey, *A first course in real analysis* (Springer)

Detailed Syllabus

1. Basic properties of $\mathbb{R}$.
2. Sequences: convergence/limits, limit laws, sandwich theorem, boundedness, monotonicity.
3. Series, ‘telescopic’ series, standard examples, tests of convergence.
4. Limits and continuity: sequential and ϵ - δ definition of limit, limit laws, sandwich theorem, limits and composition of functions, Intermediate Value Theorem, Extreme Value Theorem.
5. Differentiability: definition, differentiation rules, examples of non-differentiable functions

6. Mean Value Theorem: Mean Value Theorem, Rolle's Theorem, and theorems about monotonicity, extrema, and local extrema.
7. Integration: Definitions, elementary properties.
8. The Fundamental Theorem of Calculus.
9. Power series: radius of convergence, Taylor's formula, Taylor's series, power series as functions: differentiability, standard power series (exponential function, $\sin x$, $\cos x$, $\sinh x$, $\cosh x$)